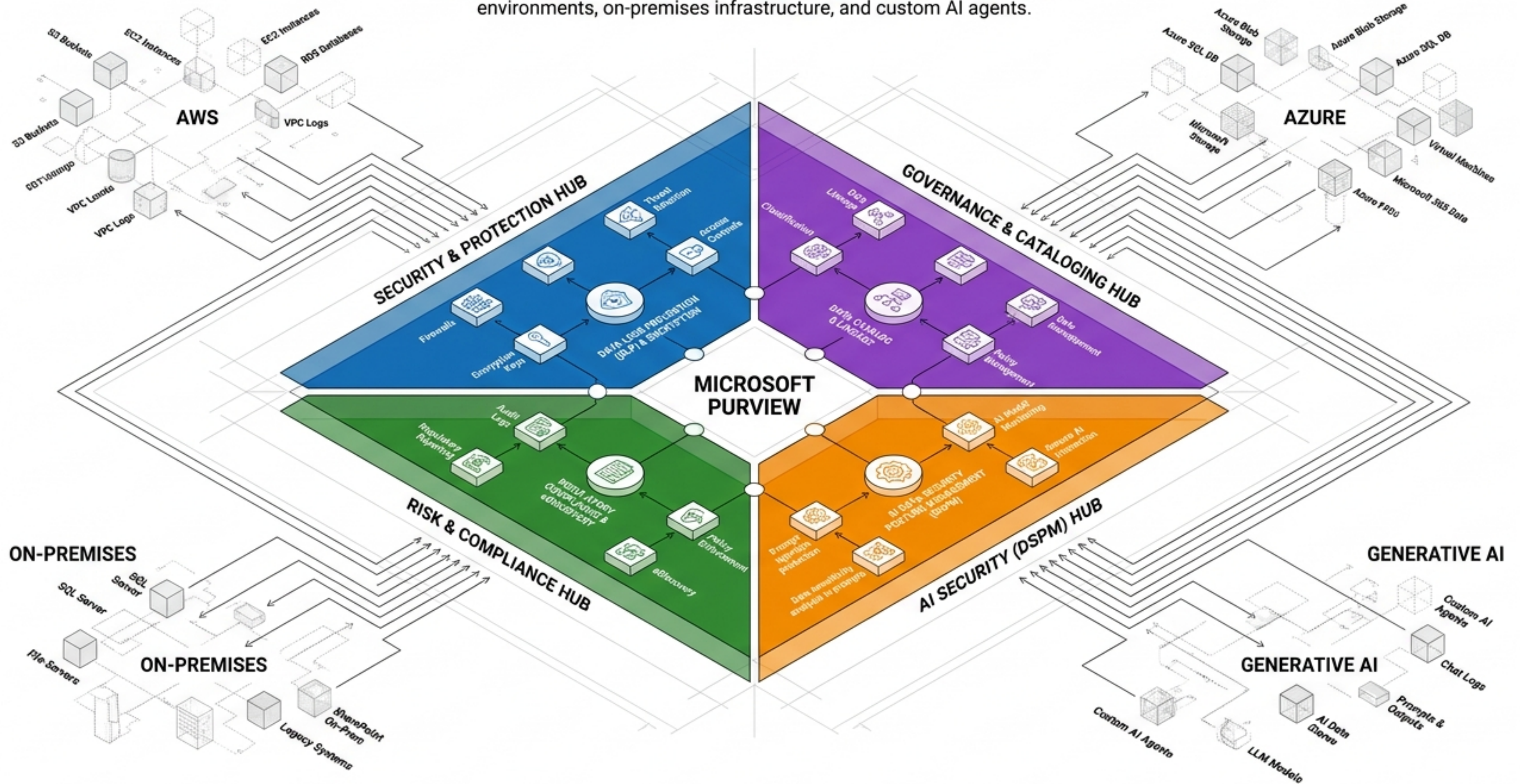
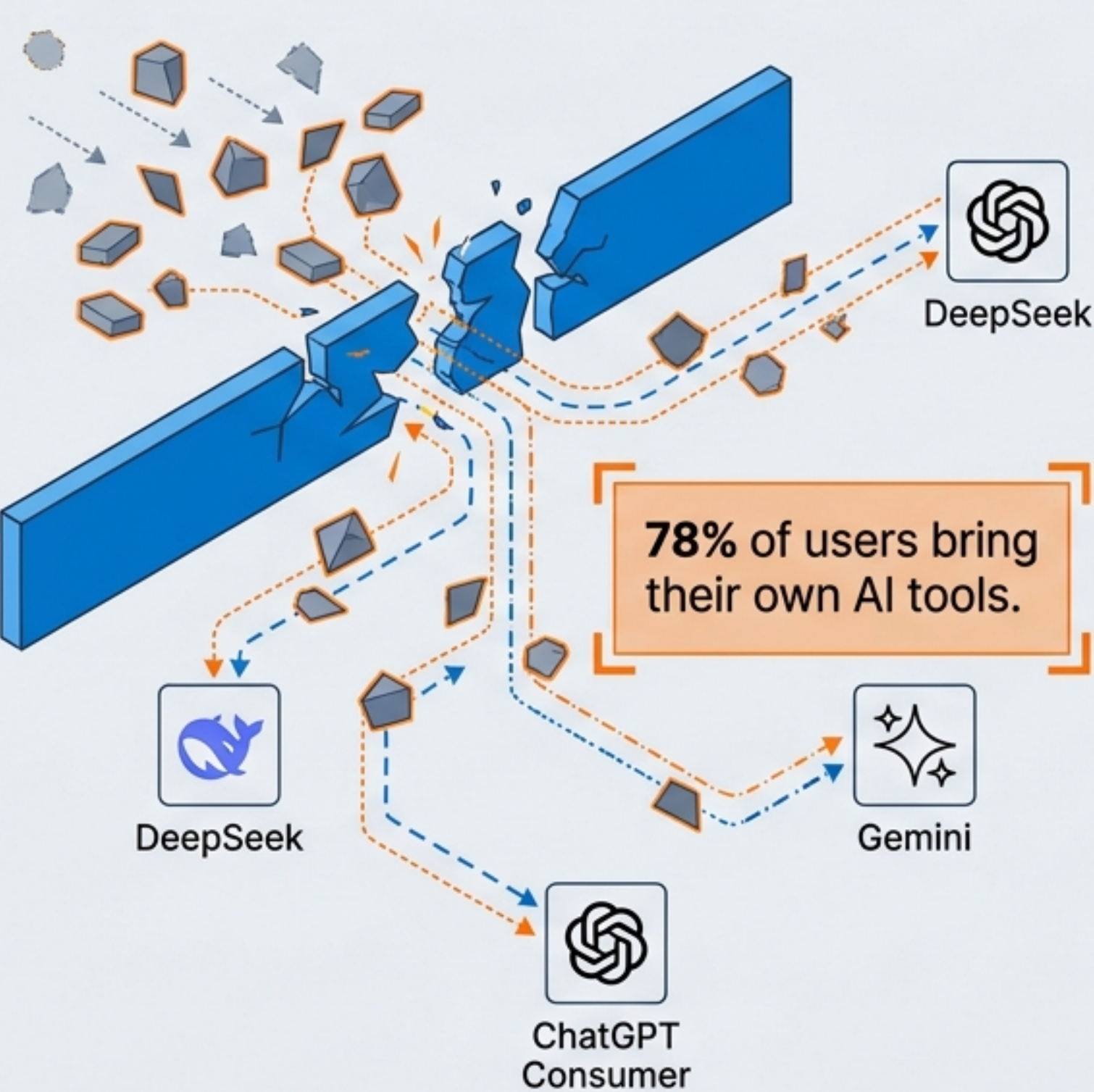


Microsoft Purview: Unified Data Security, Governance, and AI Posture

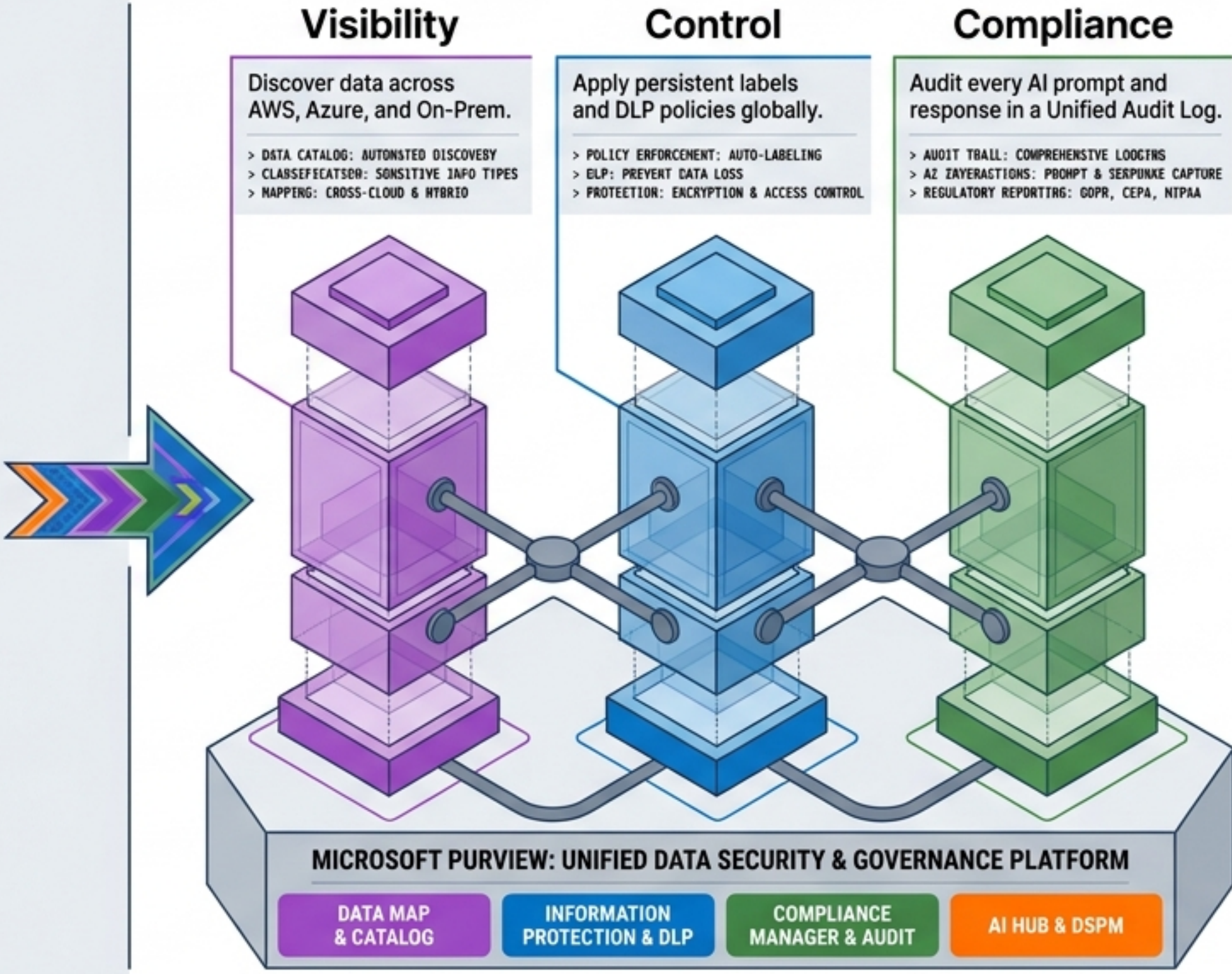
The master architectural blueprint for deploying Microsoft Purview across multi-cloud environments, on-premises infrastructure, and custom AI agents.



If we do not govern the data estate, AI adoption will accelerate sensitive data leakage.
We must unify Data Security and Governance into a single platform before scaling Microsoft 365 Copilot and custom AI agents.



UNGOVERNED FLOW: SHADOW AI & DATA EXFILTRATION RISK

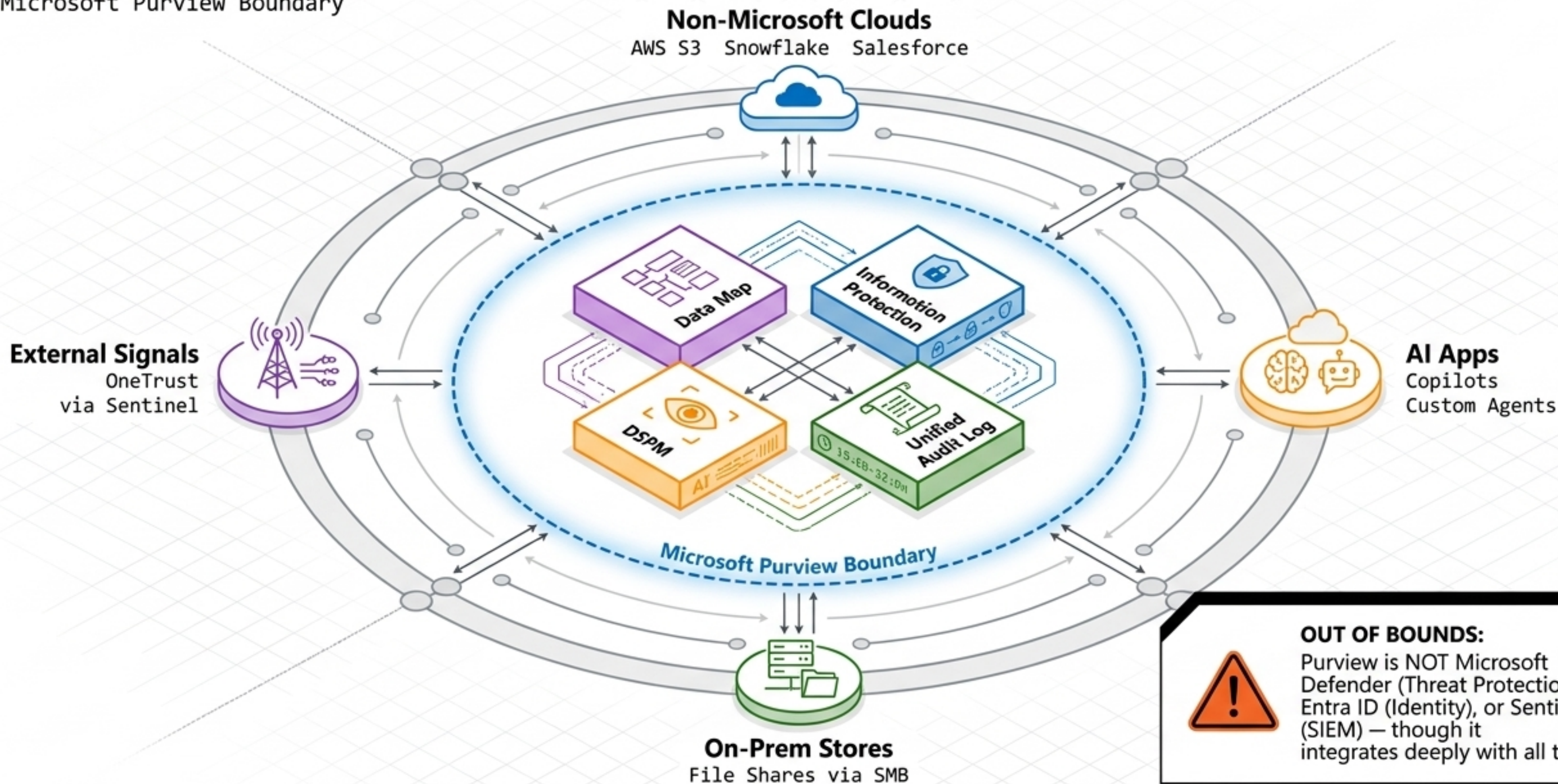


TRANSFORMED STATE: GOVERNED AI ADOPTION & SECURE DATA ESTATE

Purview sits at the center of the organization, bridging Microsoft, multi-cloud, and third-party AI.

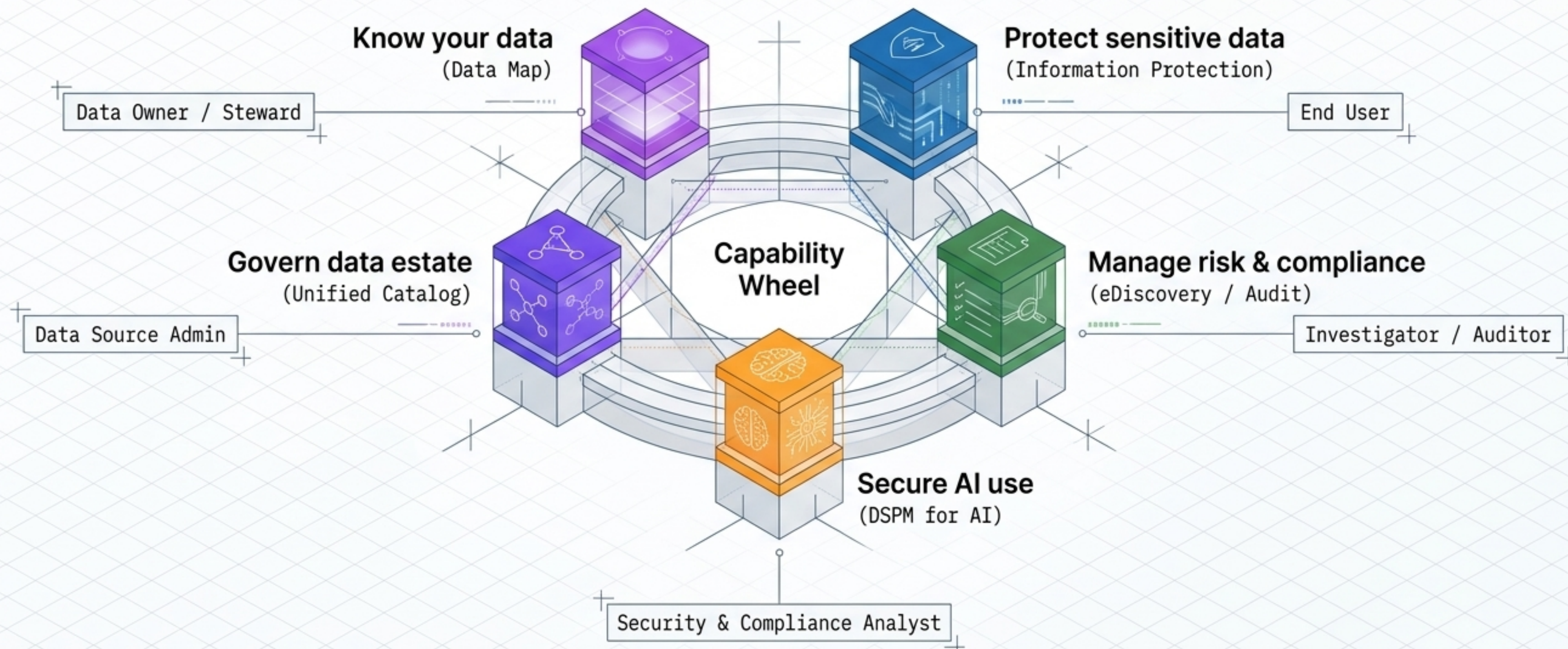
Purview extracts metadata, enforces policy, and audits activity—it does not replace your identity provider or SIEM.

● Microsoft Purview Boundary



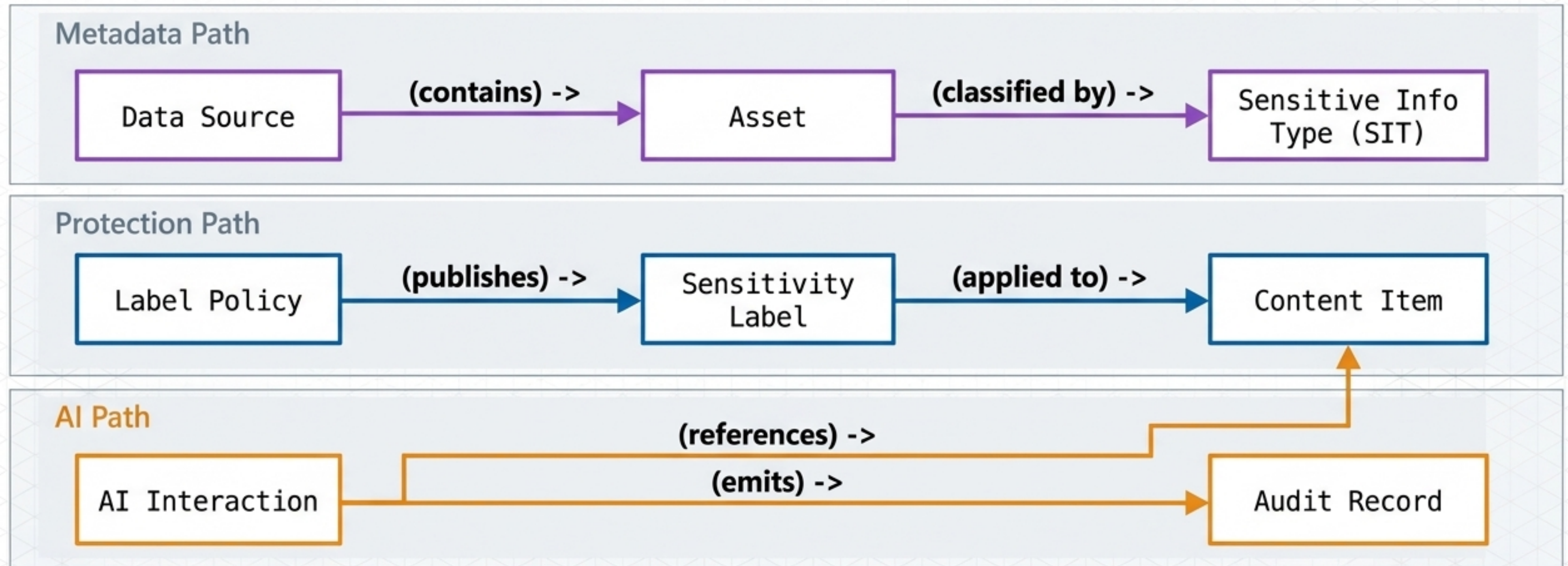
The platform delivers five core outcomes without requiring distinct, siloed products.

A single control plane enables compliance, security, and governance teams to operate on the same unified data map.

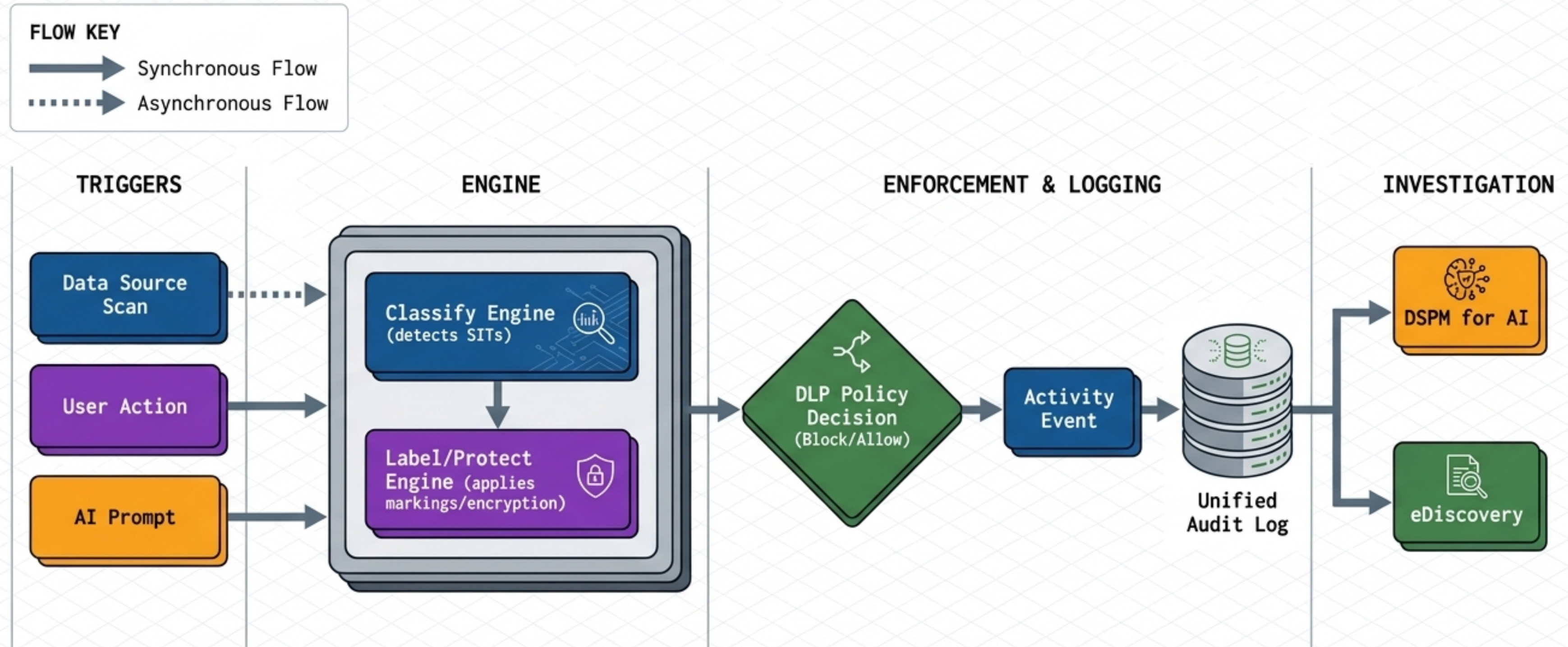


The logical model separates metadata extraction from policy enforcement and auditing.

Purview pulls metadata and classifications from the estate to build the catalog, but leaves the actual data at rest.

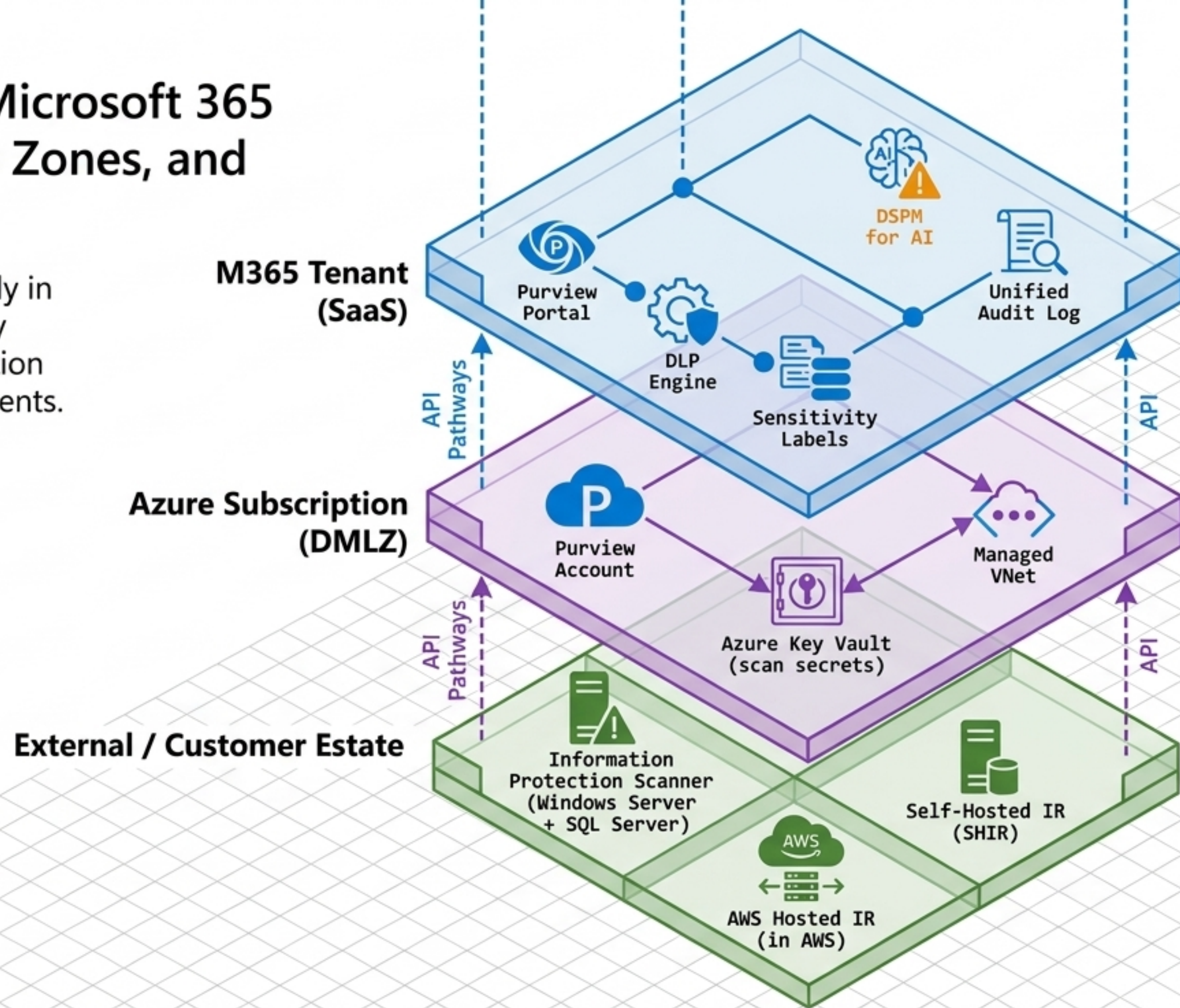


Every interaction—from a file scan to an AI prompt—triggers a synchronized evaluation path. Whether scanning an AWS bucket or querying a Copilot agent, the underlying classification and enforcement engine is identical.



Deployment spans Microsoft 365 SaaS, Azure Landing Zones, and Customer Estates

The control plane resides entirely in Microsoft's cloud, requiring only lightweight scanners or integration runtimes in your local environments.



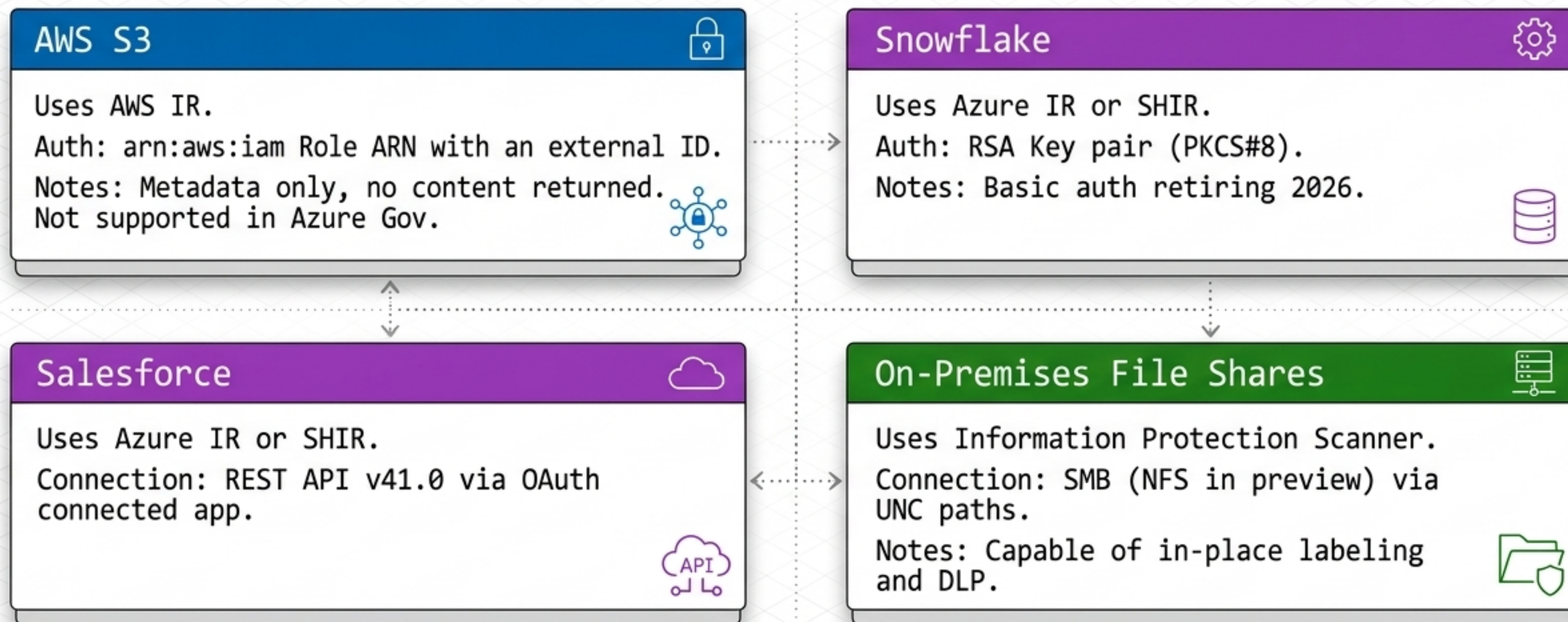
Selecting the right Integration Runtime depends entirely on network boundaries and data residency.

Default to Managed VNet IR for Azure-hosted data; deploy SHIR or Kubernetes SHIR for on-premises or strictly IP-restricted sources.

Data Source Firewall	Azure IR	Managed VNet IR	Self-Hosted IR (SHIR)	Kubernetes SHIR
Allow public access	✓	✓	✓	✓
Allow Azure/Trusted Service	✓	✓	✓	✓
Allow specific Azure VNet		✓ w/ private endpoints	✓	✓
On-premises / IP-restricted			✓ ONLY	✓ ONLY

Purview extends deep into AWS, Snowflake, Salesforce, and legacy on-premises shares.

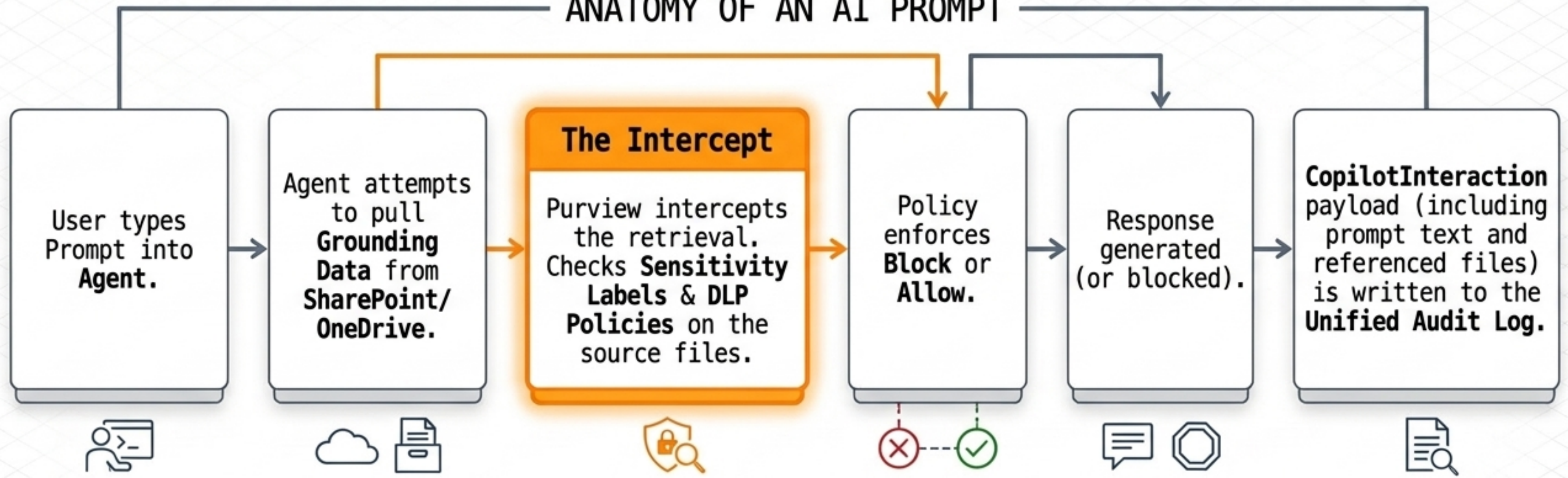
Multi-cloud scanning relies on native API integrations and IAM roles, ensuring Microsoft never requires persistent direct access to your raw data.



DSPM for AI prevents Copilots and Agents from oversharing sensitive grounding data.

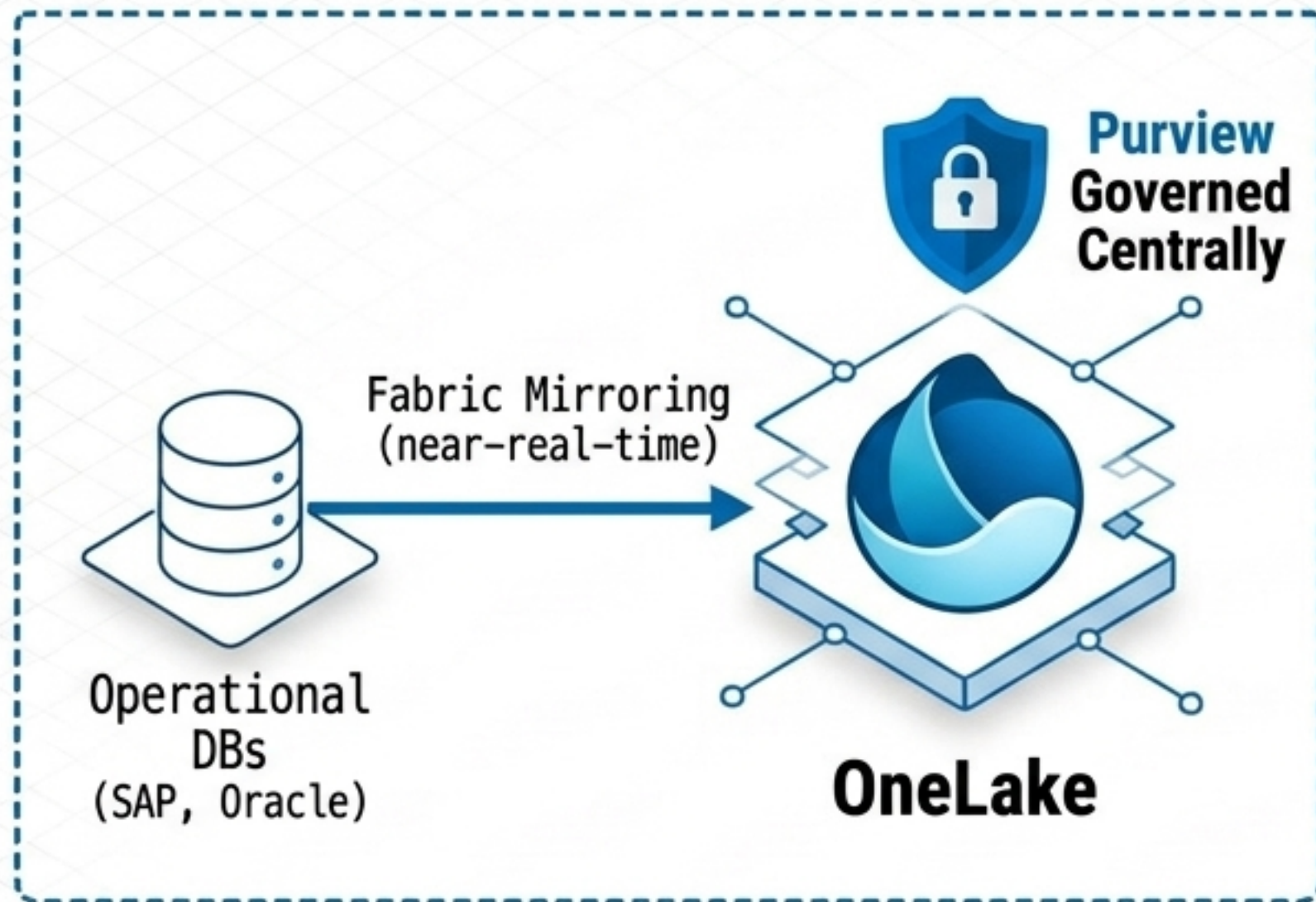
Agents inherently respect their parent application's protections; if the grounding data is **labeled and restricted**, the AI cannot bypass it.

ANATOMY OF AN AI PROMPT

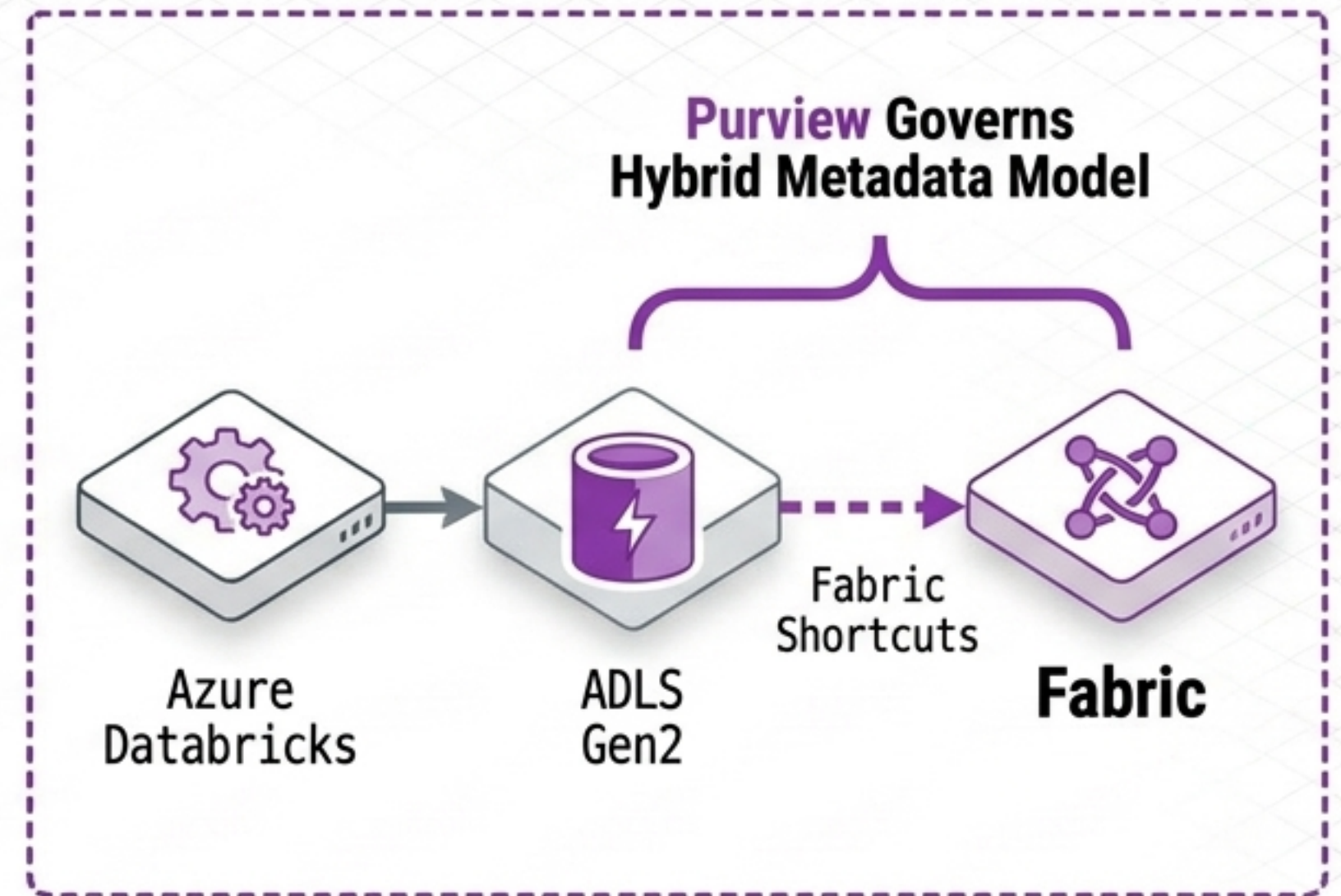


Application Landing Zones must use standardized Fabric and OneLake integration mechanisms.

Avoid direct application connections; converge enterprise data into OneLake via Fabric Mirroring to maintain a **single, governed perimeter**.



OneLake as System of Record



ADLS as System of Record

Government compliance tiers introduce strict feature gaps, particularly in multi-cloud scanning

Multi-cloud architectures in Azure Government must account for the inability to natively scan AWS S3 buckets via Purview.

Capability	Commercial	GCC (FedRAMP)	GCC High (DoD IL4)	DoD (DoD IL5)
Info Protection Scanner				
Sensitivity Labels				
DSPM for AI			Browse-to-URL unsupported	Browse-to-URL unsupported
Amazon S3 Scanning			Not Supported - Azure Gov	Not Supported - Azure Gov

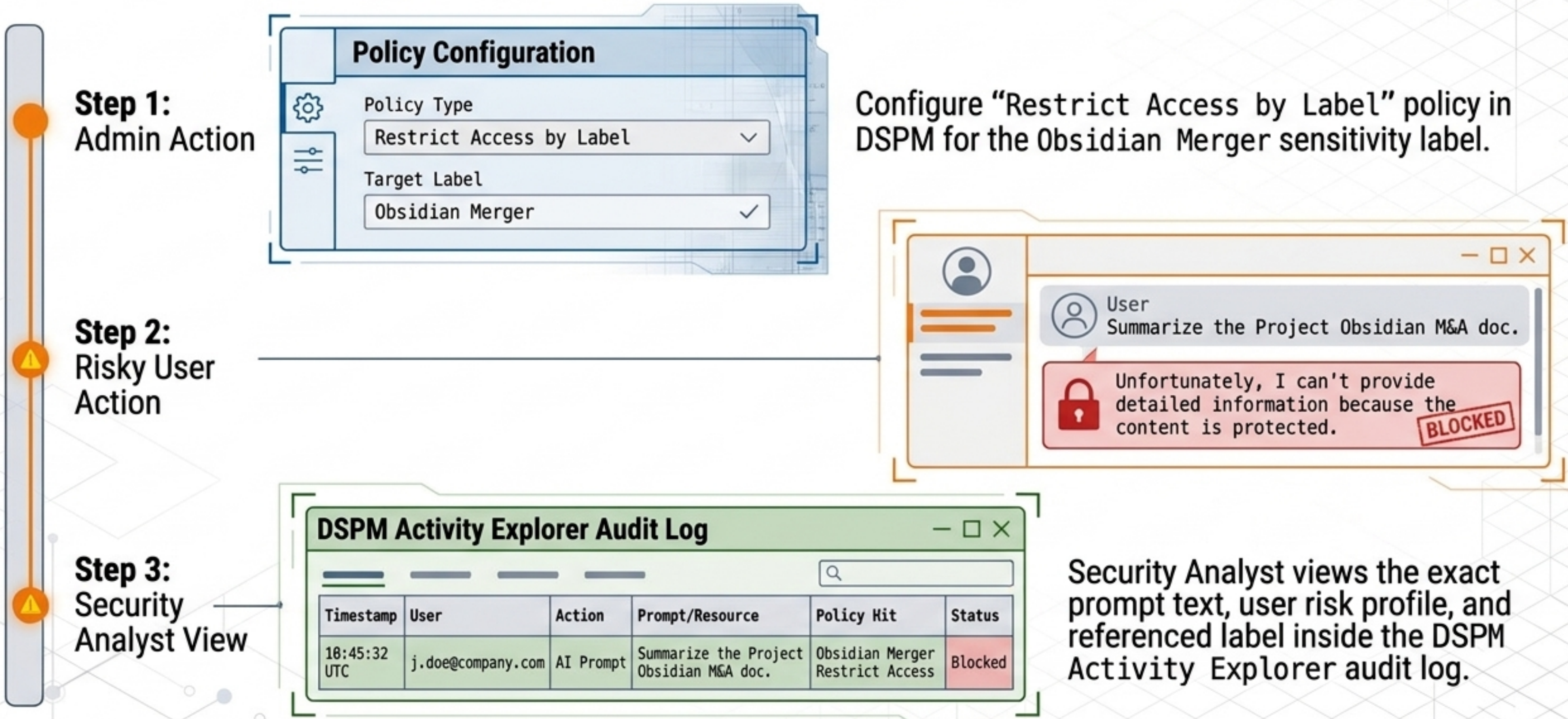
The platform is designed with defense-in-depth to catch misconfigurations and oversharing

Purview acts as a fail-safe, enforcing data boundaries even when human processes or external system configurations fail.

Threat vs. Control Failure Modes		
Failure		Control
Unlabeled sensitive M&A document dropped into an open SharePoint site.		Auto-labeling detects SITs; DLP blocks access to unauthorized users.
S3 bucket encrypted with AWS-KMS fails the metadata scan.		Administrator adds KMS Decrypt permissions to the AWS IAM Role.
Unmanaged AI app (e.g., ChatGPT Consumer) used on a corporate PC to analyze code.		Endpoint DLP and Purview browser extension block clipboard paste actions to unsanctioned URLs.

Traceability from policy creation to forensic audit is seamless and instantaneous

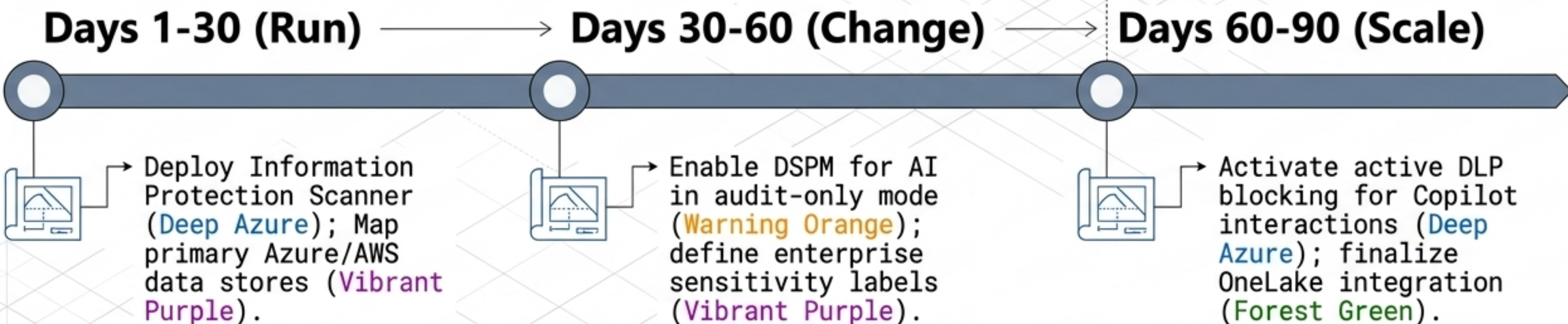
DSPM for AI transforms abstract data protection policies into **immediate, enforceable, and fully auditable AI guardrails**.



Unifying our data posture requires architecture alignment and immediate policy activation.

If we act today, we secure the Generative AI rollout by Q3.
If we delay, data fragmentation will outpace our security controls.

Next Steps



Uncertainty Register

OneTrust government-cloud support:	UNVERIFIED
FedRAMP exact certification level for GCC:	Resolve conflict